

Pranati Modumudi

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EDUCATION

Columbia University

Aug 2025 - May 2027

M.S. in Computer Science, MS Thesis Track in Machine Learning and Computational Neuroscience

Relevant Coursework: Artificial Intelligence, LLM-Based Generative AI, Computation and the Brain, Theoretical Neuroscience

University of California, Berkeley

Aug 2019 - May 2023

B.A. Data Science Honors and B.A. Economics | Regents and Chancellor's Scholar | Berkeley SCET

Relevant Coursework: Data Structures; Linear Algebra and Differential Equations; Probability & Probabilistic Modeling for Data Science; Data Engineering; Natural Language Processing; Game Theory

SKILLS

- ❖ **Programming & Tools:** Python, R, SQL, Bash, Java, JavaScript, Git, Databricks, Snowflake, Tableau
- ❖ **ML/DL:** PyTorch, TensorFlow/Keras, JAX, Scikit-learn, HuggingFace, AllenNLP, SpaCy, Neural Networks
- ❖ **LLMs & Generative AI:** Large Language Models, Prompt Engineering, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), RLHF, Model Fine-Tuning
- ❖ **Data & Research:** Experimental Design, Model Evaluation & Interpretation, Exploratory Data Analysis, Data Wrangling, ETL Pipelines, APIs, Big Data Systems

INDUSTRY EXPERIENCE

Disney Streaming, New York

Data Scientist I

Aug 2023 - May 2025

- ❖ Built end-to-end machine learning pipelines for time-series and user-level causal inference using Bayesian structural models and regression-based time-series models, enabling real-time monitoring with quantified uncertainty (80-90% confidence intervals).
- ❖ Conducted ~24 causal analyses over 2 years, evaluating business and marketing initiatives including price increases and mail campaigns, directly informing strategic decisions across Disney+ and Hulu.
- ❖ Packaged reusable Python functions via PyPi and Jenkins, adopted by five teams to standardize previously ad-hoc analyses.

OctoML, Remote

Data Science Intern

Oct 2022 - Dec 2022

- ❖ Optimized TVM machine learning compilation software by running ~100 experiments on different-sized convolution models, tuning hyperparameters to achieve ~10% $\pm$ change in convergence time.

Machine Learning Systems Intern

Mar 2022 - Jun 2022

- ❖ Diagnosed the causes of performance variance generated by auto-tuning models on machine learning compilation software TVM, as part of the Automation team.
- ❖ Published [paper](#), which was accepted to the MLArchSys International Symposium on Computer Architecture 2022.

RESEARCH/PROJECTS

Human Decision Modeling | Laboratory for Intelligent Imaging and Neural Computing, Columbia University

- ❖ Investigating how human physiological patterns (EEG signals and eye patterns) during risky and ambiguous decisions can inform artificial intelligence agents for prediction, reward shaping, and uncertainty estimation.

Multi-Agent SLM Evaluators for AI Slop Detection | AWS Small Language Build Day Hackathon

- Built an ensemble of four specialized small language models (1-3B parameters) fine-tuned with LoRA on AWS Trainium to detect low-fidelity model outputs, achieving 10 $\times$ cost reduction vs. large-model inference for real-time content quality monitoring.

AI Capability Terrain Map | Apart Research AI Forecasting Hackathon

- ❖ Developed an interactive 3D early warning system visualizing AI capabilities with real-time Epoch AI benchmark data, logistic growth modeling, and Monte Carlo forecasting across 30+ domains to identify capability trajectories, sinkholes, and emerging risks.

Perceptrons and Neural Representations | Neuromatch Academy NeuroAI Project

- ❖ Comparative analysis of spiking neural networks vs CNNs in visual working memory tasks, examining performance robustness and biological plausibility using MNIST datasets.

Undergraduate Senior Honors Thesis | Redwood Center for Theoretical Neuroscience UC Berkeley

- ❖ Selected for 15-student cohort; extended hippocampal place cell modeling from rats to bats using complex-valued neural networks to capture spatial coding differences across species under Prof. Friedrich Sommer and Dr. Gautam Agarwal.